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COMMITTED TO
RESEARCH,
INNOVATION AND
EDUCATION

47
years

Robotics and Drones Lab Project Seminar on Alcohol Detector using Arduino UNO

Core branch: Computer
Science Engineering

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Contents

- ❑ Background & Motivation – inception of the problem through time or describe the idea behind choosing this project through literature/sources
- ❑ Formulation – which explains about the methodology (circuit) followed by the platform/interface (code) used
- ❑ Results & Discussion – initial optimizations and final solutions to be discussed in detail
- ❑ Summary – bullet point representation of the project outcome/s either achieved or to be achieved
- ❑ Future scope – far sight of the chosen problem
- ❑ References

Background & Motivation

- > Drunk driving causes major accidents worldwide
- > Need for real-time detection system
- > Arduino provides low-cost solution
- > Improves road safety

Background & Motivation (Contd.,)

- Increasing need for smart safety systems
- Alcohol sensors detect ethanol in breath
- Can prevent accidents using alerts
- Prototype safety solution

Formulation

- The formulation of the Alcohol Detection System involves designing both hardware and software components to detect alcohol concentration and generate alerts.
- 1. System Design
- The system is built using Arduino UNO as the main controller.
- An MQ-3 Alcohol Sensor is used to detect alcohol levels in the air.
- Output devices like a buzzer and LED are used for alerts.
- 2. Hardware Implementation
- MQ-3 sensor is connected to the analog input pin (A0) of Arduino.
- LED and buzzer are connected to digital pins.
- Proper power supply (5V) and grounding are ensured.

- 3. Software Implementation
- Arduino is programmed using Arduino IDE.
- The program:
- Reads analog values from the sensor
- Converts them into digital readings
- Compares with a predefined threshold
- 4. Working Logic
- If alcohol concentration $<$ threshold:
- System remains normal (LED OFF, buzzer OFF)
- If alcohol concentration $\geq$ threshold:
- LED turns ON
- Buzzer sounds alert

- 5. Flow of Operation
- Sensor detects alcohol vapor
- Signal sent to Arduino
- Arduino processes data
- Decision based on threshold
- Alert is triggered
- 6. Key Considerations
- Sensor calibration is required for accuracy
- Threshold value must be properly tuned
- Environmental factors may affect readings

Results & Discussion

- Detects alcohol successfully
- Fast response time
- Sensitivity depends on calibration

Results & Discussion (Contd.,)

- Advantages: Low cost, portable
- Limitations: Environmental interference
- Requires calibration

Summary

- Implemented alcohol detection system
- Real-time alert mechanism
- Useful for road safety

Future scope

- Engine lock system
- IoT integration
- GSM alerts
- Advanced sensors

References

- Arduino Documentation
- MQ-3 Datasheet
- Research papers on alcohol detection